ACF · Capital investment appraisal — Question Patterns and Methods

Based on 24 past-exam parts, 2020–2025, their marking keys and the examiners' reports.

First: where marks are lost most

- Discounting the wrong year or leaving a flow out of the present value. Keys award marks per year for the correct discount factor, and for the initial outlay discounted at time zero. Students who apply one factor to all years lose several marks at once (2024 SA Q17(a), 2020 SA Q16(b)). The key also credits placing the second $30,000 computer payment in Year 1 rather than in the initial cost (2025 SA Q21(b)).
- Leaving residual or salvage value out of the cash flows. Keys give a separate mark for identifying the residual value and another for including it in the year it is received (2024 SA Q17(a), 2023 SA Q16(a), 2020 SA Q16(b)).
- Keeping non-cash depreciation in the cash flows, or omitting capitalised costs from the depreciable base. Keys reward removing depreciation and adding installation costs to the depreciable amount before calculating annual cash flows (2025 SA Q21(a)).
- Payback answers that are not converted into the requested units. Keys credit the unrecovered amount, the fraction of the next year's inflow, and the conversion to months, and then a final statement in whole years and months (2024 SA Q17(b), 2021 SA Q20(b)).
- Recommendations that are stated but not justified from the workings. Keys give one award for the decision and a separate award for the justification, which must refer to the NPV, payback or the comparison of options (2025 SA Q21(c), 2024 SA Q17(d), 2023 SA Q16(b)).
- Non-financial factors named but not explained. Examiners' keys give 1 mark for a general comment and more for a described effect on the specific decision (2025 EA Q23(a), 2022 EA Q22(b)). The examiners' reports for this topic are brief; this point rests on the marking keys.

Pattern 1: Estimate relevant investment cash flows

How often: 4/6 papers · 10.0 marks when it appears · 2.9% of a paper.

Recognise it: a project described by an up-front cost, installation or further instalments, a useful life, operating costs, receipts, and a tax rate. The question asks for annual net cash flows or the cash flows that feed a later NPV or payback calculation.

Method

1. Note the cost, any installation or extra instalment, the useful life, any residual or salvage value, the tax rate and the rounding instruction.
2. Work out the depreciable amount: the cost plus capitalised extras, less any residual value.
3. Calculate straight-line depreciation over the useful life. Depreciation is a working figure only.
4. Calculate cash receipts less cash operating costs for each year.
5. Adjust for tax: take the after-tax effect of the taxable profit (receipts less operating costs less depreciation), then add back the non-cash depreciation so that only cash remains.
6. Place any capital outlay at its correct time: the initial cost at Year 0, and any later instalment in the year it is paid.
7. Show each year in a table, with outflows negative, and keep the rounding stated in the question.

What the key rewards:

- the depreciable base including installation;
- the correct life used in straight-line depreciation;
- the operating expenses deducted in the annual cash flow;

- the resulting after-tax annual net cash inflow;
- for the NPV part, the first and second computer payments placed in the correct years;
- the installation cost included in the project cost.

Common errors: keeping depreciation in the cash flow; leaving installation out of the depreciable base; omitting the second instalment of the computer payment; leaving out the year 5 salvage or residual inflow; using the wrong life.

Past questions: 2025 SA Q21(a), 2025 SA Q21(b), 2024 SA Q17(a), 2024 SA Q17(b), 2023 SA Q16(a), 2020 SA Q16(a), 2020 SA Q16(b)

Pattern 2: Calculate the payback period

How often: 4/6 papers · 3.0 marks when it appears · 0.9% of a paper.

Recognise it: the question asks for the "payback period" or "how long to recover" the investment, usually with a table of annual net cash flows and a single initial cost, sometimes with a required period to compare against.

Method

1. Write the initial cost as a positive figure.
2. Accumulate the annual net cash inflows year by year until the running total reaches or passes the cost.
3. Find the amount still unrecovered at the end of the last full year.
4. Divide the unrecovered amount by the next year's inflow to find the fraction of that year.
5. Convert the fraction into the requested units (usually months, using 12 months per year).
6. Round as the question requires and state the answer in years and months.
7. Check the answer against the cumulative figures: the period should fall between the year in which recovery occurs and the next.

What the key rewards:

- the accumulated inflows at the end of each year;
- the unrecovered investment as cost less accumulated inflows;
- the fraction taken from the next year's inflow;
- the conversion to months;
- the final period stated in whole years and months;
- where the question gives several initial-cost components, each component is awarded separately.

Common errors: dividing by the wrong year's inflow; forgetting to convert the fraction to months; rounding the months incorrectly or reporting a figure that does not match the fraction; using the total of all inflows rather than the amount still unrecovered. Multiple-choice versions (2022 MC Q5) require picking the option that matches the correct calculation.

Past questions: 2024 SA Q17(b), 2022 MC Q5, 2021 SA Q20(b), 2020 SA Q16(a)

Pattern 3: Discount project cash flows and calculate NPV

How often: 5/6 papers · 5.4 marks when it appears · 2.0% of a paper.

Recognise it: a request for the "net present value" of a project, with a discount rate given and cash flows listed by year. Sometimes an annuity or a single-sum factor is given in the question or tables.

Method

1. Set out the cash flows by year in a table, with the initial outlay at Year 0 and each later flow in its year.
2. Look up the discount factor for each year at the given rate; the Year 0 factor is 1.
3. For an even annual flow across several years, use the annuity factor for the whole block rather than each year separately, and check that the block is correctly positioned in time.
4. Multiply each flow by its factor to get its present value; include any residual or salvage value in the

year it is received.
5. Discount any later outlay or instalment at the factor for the year in which it occurs, reading its timing from the question.
6. Add the present values of inflows, subtract the present values of outflows, and state the NPV.
7. Round as instructed, usually to the nearest dollar.

What the key rewards:

- the correct annual net flow used in an annuity;
- the annuity factor applied to later flows;
- the single-sum factor applied to the Year 1 payment or to a later cost;
- the present value of inflows and of outflows as separate totals;
- the final NPV; each year's discounted flow is credited separately.

Common errors: applying the year 1 factor to every year; discounting a later payment at the wrong factor; leaving the residual value out of the present value; subtracting the outflows before discounting; using the undiscounted total.

Past questions: 2025 SA Q21(b), 2024 SA Q17(a), 2023 SA Q16(a), 2021 SA Q20(a), 2020 SA Q16(b)

Pattern 4: Recommend an investment using appraisal results

How often: 6/6 papers · 3.7 marks when it appears · 1.6% of a paper.

Recognise it: a request to "state whether" a proposal should be accepted or rejected, which option to choose, or which location to recommend, after NPV or payback results have been calculated. The word "justify" or "with reference to your calculations" signals that a reason is needed.

Method

1. Read the decision question carefully: accept or reject a single project, choose between options, or choose between locations.
2. Take the NPV and payback results from your workings, and state the decision in one clear sentence.
3. Quote the NPV from your workings as the evidence for the decision; a positive NPV is itself a reason to proceed (2025 SA Q21(c)).
4. Give the payback result from your workings as a further reason, quoting the period.
5. For a choice between options, compare the NPVs and paybacks from your workings and name the option the comparison supports; the 2020 SA Q16(c) key rewards explaining the choice by the higher NPV.
6. Give at least one reason for the decision, quoting the figures from your workings, and a second reason if the question asks for two.

What the key rewards:

- the decision itself (proceed or reject, or which option or location);
- a justification that refers to the NPV, the present value compared with cost, the payback period, or the comparison of the options;
- a second reason where two are asked;
- no award for a decision with no link to the calculations.

Common errors: stating a decision without any reason; giving a reason that does not come from the calculations; comparing only one measure when the question asks for both; recommending the option with the higher payback where NPV points the other way without explaining why.

Past questions: 2025 SA Q21(c), 2024 SA Q17(a), 2024 SA Q17(b), 2024 SA Q17(d), 2023 SA Q16(b), 2022 EA Q22(a), 2021 SA Q20(c), 2020 SA Q16(c)

Pattern 5: Evaluate non-financial investment factors

How often: 4/6 papers · 3.5 marks when it appears · 1.0% of a paper.

Recognise it: a request to "describe" or "discuss" non-financial factors affecting a capital investment, such as government regulation, competition, location, consumer preferences, or a machine's other effects. These are usually asked in extended answers and in the "describe one or two" parts.

Method

1. Identify the factor named in the question, or choose a relevant one from the stimulus material.
2. Say what the factor is in a sentence.
3. Explain how it affects the specific decision in the question (for example, how a regulation changes the cost or the timing of a factory opening).
4. Link the effect to the case: refer to the locations, products or numbers in the stimulus.
5. Repeat for the second factor if two are asked.

What the key rewards:

- for each factor, the identification;
- a description linked to the decision;
- a general comment with no link to the case earns only 1 mark in the extended question; the rest needs a description or explanation tied to the decision.

Common errors: naming a factor without describing its effect; writing a generic comment with no link to the case; repeating one factor under two headings; describing a financial factor (such as NPV) as if it were non-financial.

Past questions: 2025 EA Q23(a), 2024 SA Q17(c), 2022 EA Q22(b), 2021 SA Q20(d)

Pattern 6: Explain capital investment and time value of money

How often: 4/6 papers · 2.5 marks when it appears · 0.7% of a paper.

Recognise it: a request to "explain" or "describe" the nature of capital investment decisions, or a multiple-choice item about the time value of money or the effect of a change in the discount rate on NPV or payback.

Method

1. For an explanation or description: state that capital investment is a long-term commitment of large funds to long-lived assets.
2. Give at least two examples of such assets, other than the one in the question where the question so requires.
3. For the time value of money: remember that money received later is worth less today, so future flows are discounted.
4. For a change in the rate: a higher discount rate lowers the NPV; a lower rate raises it. Payback does not use the discount rate, so it is unaffected.
5. For a multiple-choice item, eliminate options that confuse the direction of the effect or the time scale.

What the key rewards:

- for a description, the nature of capital investment (large, long-term, asset-based) and the two examples asked for;
- for the time-value item, the option stating that money's value declines over time;
- for the discount-rate item, the option in which NPV moves and payback is unchanged;
- multiple-choice items award one mark for the letter.

Common errors: describing capital investment as short-term; giving examples that are not long-lived assets; reversing the direction of the discount-rate effect on NPV; saying payback changes when the rate changes.

Past questions: 2025 EA Q23(a), 2024 MC Q5, 2024 MC Q10, 2022 MC Q9, 2020 SA Q16(d)